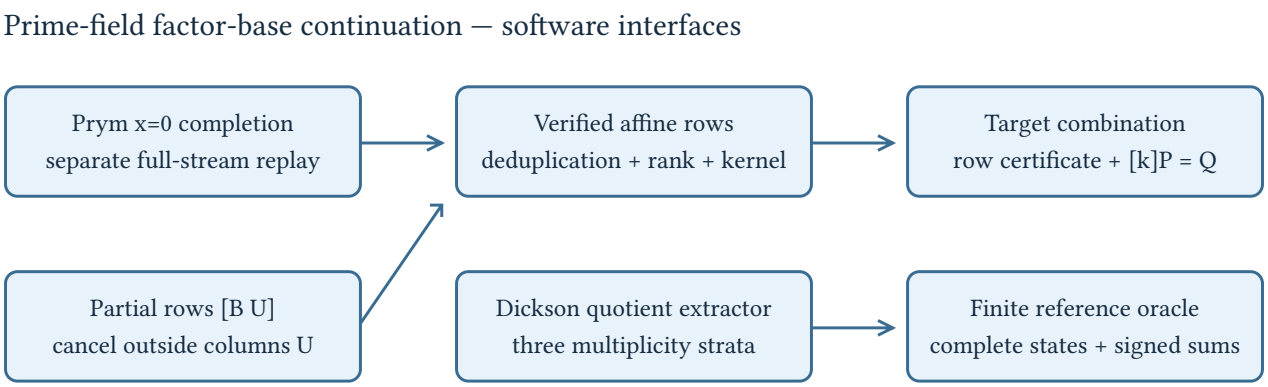
Prime-field factor-base continuation

Software implementation and validation · 28 October 2026

This delivery implements four follow-ups from the repository survey. It is not a scientific experiment, an independent review, or a speedup claim.



No new scientific panel, performance measurement or research-status transition is represented.

Implemented

- Exact affine relation diagnostics: original-row certificates, affine-equation deduplication, complete kernels and deficient-rank target identification.
- Partial-relation cancellation: eliminate outside points before counting new information on the existing factor base. Every equation retains its weights.
- Additive Prym x=0 source/interpolation completion, with full-stream replay in the separate arithmetic lane; the old frozen collector is unchanged.
- Complete Dickson extractor driver: all three multiplicity strata, finite resultant and signed point-sum oracle, resource guards and arity-four controls.

Validation scope

The tests use small software fixtures: the exterior extractor is compared with complete finite sets at depths 2–4 over p=127, while Prym zero completion is checked over p=101. The affine diagnostics are checked against every solution of small systems. Deliberate missing solutions, false signs, inconsistent rows, false digests/ranks and unadmitted collection are rejected. See `validation.json` for exact commands, initial failures and final outcomes.

No P-256 fixed-panel execution or p=4099 collector trial was launched. These remain separate frozen-protocol, native-supervision and checkpoint operations. The entry points bind committed source/input bytes and call the existing real checkpoint gate; they provide no bypass or execution release.

Sources and limits

Petit–Kosters–Messeng, PKC 2016, sections 2.4 and 3.1–3.5, retrieved and read: primary paper. The composed-map construction is prior work, and the relation solver and linear algebra must both be charged. Small input degree is not a cost bound.

Reused repository source: the October 26 F1 core/exterior implementation and the October 19/25 Prym producer, Census and independent verifier at main base `cdd2150922211f180ab3d8c3f3b248356c78f025`. The accompanying README records individual paths and the issue-only October 27 evidence gap.

The exterior dimension remains binomial in fibre size and exponential in depth. Individual factor-base logs need not be identifiable for one combination to be recoverable, but a cheap natural target decomposition is still required. No canonical scientific metric, hypothesis status or historical run is changed.